UltraGroth. Lookup Checks Enabled in Groth16

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Distributed Lab

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Introduction

Motivation

We typically want to check **inclusion** $\{z_i\}_{i\in[n]} \subseteq \{t_j\}_{j\in[v]}$, where $\{z_i\}_{i\in[n]}$ is part of the witness while $\{t_j\}_{j\in[v]}$ is the lookup table.

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Example usage: effective range-checks (Bionetta, Rarimo circuits, non-native ZK verifications etc.)

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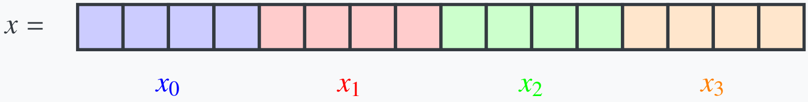
16 constraints



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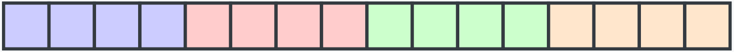
$x =$ 

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Generally: for n-bit range-check, the circuit's complexity reduces from $\mathcal{O}(n)$ to $\mathcal{O}(2^w + \frac{n}{w})$, which yields $\mathcal{O}(n/\log n)$ assymptotic.

Logup Check

Theorem (Some stuff from ZKDL Camp)

The inclusion check $\{z_i\}_{i \in [n]} \subseteq \{t_i\}_{i \in [v]}$ is satisfied if and only if there exists the set of multiplicities $\{\mu_i\}_{i \in [v]}$ where $\mu_i = \#\{j \in [n] : z_j = t_i\}$ such that for $\gamma \leftarrow\$ \mathbb{F}$:

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Naive approach: Define `signal` `gamma` and implement this check in-circuit. This costs exactly $n + 2v$ constraints.

Problem: We *cannot define random* signals in Circom since Groth16, compared to $\mathcal{P}\text{lon}\mathcal{K}$ or sumcheck-based approaches, is not compiled from interactive protocol using Fiat-Shamir heuristic.

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We can instead apply the **Freiveld's protocol**. Sample random $\gamma \leftarrow\$ \mathbb{F}^n$, compute C off-circuit and then verify:

$$AB\gamma = C\gamma \qquad \texttt{// Costs } 3n^2 \texttt{ constraints}$$

Example: Attention layer implementation in zkML.

Plan

1. We recap the Groth16 construction.
2. We identify how to make it interactive.
3. We specify how Fiat-Shamir transformation should be applied.
4. We show how it can be practically implemented.

Groth16 Recap

R1CS

Recall that we can encode any NP statement in the form of m equations of form $\langle \ell_j, \mathbf{z} \rangle \cdot \langle \boldsymbol{r}_j, \mathbf{z} \rangle = \langle \boldsymbol{o}_j, \mathbf{z} \rangle$ for $j \in [m]$ and $\mathbf{z} \in \mathbb{F}^n$. Such way of representing the statement is called **R1CS arithmetization**.

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- ✓ As of now, this is one of the most optimal arithmetization systems available (compared to PlonK and AIR).
- ✓ All linear operations over elements of $\mathbf{z}$ cost **0 constraints**, compared to PlonK.

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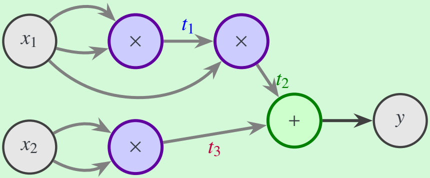
Circuit Diagram



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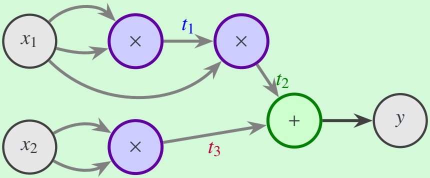

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In this case, the witness looks as $z = (1, x_1, x_2, t_1, t_2, t_3, y)$, and (for simplicity, consider only L, R):

$$L = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

QAP

We interpolate the columns of each of the matrices, thus getting $3n$ polynomials $\{(\ell_i(X), r_i(X), o_i(X))\}_{i\in[n]} \subseteq \mathbb{F}^{\leq m}[X]$:

$$\ell_i(\omega^j) = L_{i,j}, \quad r_i(\omega^j) = R_{i,j}, \quad o_i(\omega^j) = O_{i,j}, \quad i \in [n],\ j \in [m]$$

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$$\sum_{i\in[n]} z_i\ell_i(X) \cdot \sum_{i\in[n]} z_i r_i(X) = \sum_{i\in[n]} z_i o_i(X) + t_\Omega(X)h(X),$$

where $h(X)$ is computed by a prover and $t_\Omega(X) \triangleq \prod_{h\in\Omega}(X - h)$ is the *vanishing polynomial* over *evaluation domain* $\Omega = \{\omega^j\}_{j\in[m]}$. The corresponding relation:

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$$\mathcal{R}_{\mathsf{QAP}} = \left\{ \begin{matrix} \mathbb{x} = \{z_i\}_{i\in\mathcal{I}_X} \\ \mathbb{w} = \{z_i\}_{i\in\mathcal{I}_W} \end{matrix} \;\middle|\; \begin{matrix} \sum_{i\in[n]} z_i\ell_i(X) \cdot \sum_{i\in[n]} z_i r_i(X) = \sum_{i\in[n]} z_i o_i(X) + t_\Omega(X)h(X) \\ \text{for some } h(X) \in \mathbb{F}[X] \end{matrix} \right\}$$

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- Verify$(\sigma, \mathbb{x}, \pi) \to \{0, 1\}$. The verifier gets the arithmetic circuit $t : \mathbb{F}^{m+k} \to \mathbb{F}^\eta$ of degree d and verifies whether $t(\sigma, \pi) = 0$.

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Groth16 is essentially a Linear NIP where $d = 2$ and σ is given by:

$$\sigma = \left(\alpha, \beta, \gamma, \delta, \{\tau^i\}_{i\in[n]}, \left\{\frac{\zeta_i(\tau)}{\gamma}\right\}_{i\in[m]}, \left\{\frac{\tau^i t_\Omega(\tau)}{\delta}\right\}_{i\in[n]}\right),$$

with $\zeta_i(X) := \beta\ell_i(X) + \alpha r_i(X) + o_i(X)$ and $\tau = (\alpha, \beta, \gamma, \delta, \tau)$.

Groth16 Construction

Fix bilinear group $\mathcal{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e)$ with pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$.

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$$a(X) = \alpha + \sum_{i \in [n]} z_i \ell_i(X) + r\delta, \;\; b(X) = \beta + \sum_{i \in [n]} z_i r_i(X) + s\delta,$$

$$c(X) = \delta^{-1} \left(\sum_{i \in \mathcal{I}_W} z_i \zeta_i(X) + h(X) t_{\Omega}(X) \right) + a(X)s + b(X)r - rs\delta$$

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- $\mathsf{Verify}(\mathsf{vp}, \mathbb{x}, \pi) \to \{0, 1\}$. Parse $\pi = (\pi_A, \pi_C, \pi_B)$ and accept the proof if and only if

$$e(\pi_A, \pi_B) = e(g_1^\alpha, g_2^\beta) \cdot e(g_1^{\iota(\tau)}, g_2^\gamma) \cdot e(\pi_C, g_2^\delta),$$

where $\iota(X) := \gamma^{-1} \sum_{i \in \mathcal{I}_X} z_i \zeta_i(X)$ is the input commitment.

UltraGroth

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Problem. We cannot practically “hash” the witness part $\mathbb{w}_0$.

2-round UltraGroth Construction

Split public indexing set $\mathcal{I}_X$ into two parts: $\mathcal{I}_X^{\langle 0\rangle}$ and $\mathcal{I}_X^{\langle 1\rangle}$. Similarly, split the witness indexing set $\mathcal{I}_W$ into $\mathcal{I}_W^{\langle 0\rangle}$ and $\mathcal{I}_W^{\langle 1\rangle}$.

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Conclusion

UltraGroth protocol's verifier is only **4 pairings**, **1 hashing operation**, and $\mathcal{O}(|\mathbb{x}|)$ exponentiations over $\mathbb{G}_1$.

Multi-round UltraGroth

Definition (dQAP)

We define the $(d+1)$**-round quadratic arithmetic program** (or d**QAP**, for short), as follows:

$$\mathcal{R}_{\mathsf{dQAP}} = \left\{ \begin{array}{l} \mathbb{x}_i = \{z_j\}_{j \in \mathcal{I}_X^{\langle i \rangle}} \\ \mathbb{w}_i = \{z_j\}_{j \in \mathcal{I}_W^{\langle i \rangle}} \\ \text{for } i \in [d+1] \end{array} \middle| \begin{array}{c} \ell(X) \cdot r(X) = o(X) + t_\Omega(X) h(X) \\ \ell(X) = \sum_{i \in [n]} z_i \ell_i(X), \\ r(X) = \sum_{i \in [n]} z_i r_i(X), \\ o(X) = \sum_{i \in [n]} z_i o_i(X), \\ \text{for some } h(X) \in \mathbb{F}[X] \end{array} \right\},$$

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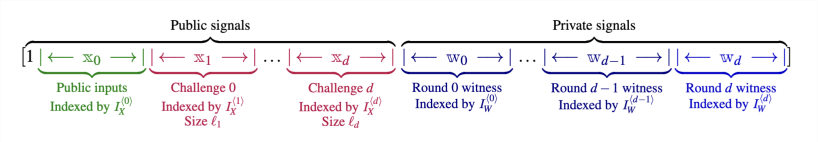

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Define **strategy** for $\mathcal{R}_{\mathsf{dQAP}}$ as the collection of functions $S = \{S_i\}_{i \in [d]}$ each of which computes the witness for the given round given previous witnesses and challenges and the current challenge, sampled by the verifier. In other words,

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Example

0QAP represents the regular QAP with the strategy $S = \{S_0\}$ that consists of the witness generator: $\mathbb{w} = S_0(\mathbb{x})$.In turn, 1QAP represents the lookup Groth16 version where $\mathbb{w}_0 = S_0(\mathbb{x}_0)$ computes the witness without lookups while $\mathbb{w}_1 = S_1(\mathbb{x}_0, \mathbb{x}_1, \mathbb{w}_0)$ computes lookup constraints.

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Initialize accumulator $a_0 := \mathcal{H}(\sigma)$.

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- If $i < d$, for each $j \in \mathcal{I}_X^{\langle i+1 \rangle}$, set $z_j \leftarrow \mathcal{H}(a_{i+1}, g_1^j)$.

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Verification: $e(\pi_A, \pi_B) = e(g_1^\alpha, g_2^\beta) \cdot e(g_1^{i(\tau)}, g_2^\gamma) \cdot \prod_{i \in [d+1]} e(\pi_C^{\langle i \rangle}, g_2^{\delta_i})$.

UltraGroth Efficiency

Groth16 performance over the circuit of size n and statement size ℓ.

- **Prover work**: MSM of size $\mathcal{O}(n)$ over $\mathbb{G}_1$ and $\mathbb{G}_2$.
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UltraGroth performance over $\mathcal{R}_{\mathsf{dQAP}}$ in turn:

- **Prover work**: MSM of size $\mathcal{O}(n/\log n)$ over $\mathbb{G}_1$ and $\mathbb{G}_2$.
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- Allowed interactiveness for potentially more complex protocols.

Any Questions?

♥

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